Avatar Sargamantha Ndoen

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Portfolio — click to view projects

Research Interests

Medical and assistive robotics; teleoperation; compliant mechanisms; actuation and transmissions; computational design; model-based control.

Education

Institut Teknologi Bandung 2021–2025
Bachelor of Engineering in Mechanical Engineering *GPA: 3.80/4.00*

Published Undergraduate Thesis

A Surrogate-Assisted Framework for the Subject-Specific Scalable Design of Compliant Prosthetic Wrists
Institut Teknologi Bandung | Supervisor: Prof. Sandro Mihradi
Scientific Reports 16, 32606 (2026). doi:10.1038/s41598-026-62052-x

- Designed a two-DoF compliant prosthetic wrist and automated SolidWorks–ANSYS analysis to generate 4,096 FEA samples.
- Used ML surrogates and multi-objective optimization to produce five personalized wrist designs; rechecked them using FEA and fabricated a prototype for range-of-motion testing.

Selected Engineering Projects

BLDC–Cycloidal Actuators and Teleoperated Arm
- Designed and built backdrivable BLDC–cycloidal actuators with a 12:1 gear ratio and a robotic arm; integrated wearable teleoperation.

Tendon-Driven Compliant Hand
- Designed and built a 15-DoF tendon-driven hand with compliant joints; demonstrated commanded hand gestures.

Tendon-Driven Continuum Robot
- Designed and built a six-bending-DoF continuum robot and compliant gripper; demonstrated needle pickup and insertion, and tested individual-DoF motion isolation and parasitic motion.

Beltless Continuously Variable Transmission
- Modeled RatioZero's beltless CVT in Python, used surrogate-assisted optimization to reduce modeled torque ripple, and built a working 3D-printed proof of concept.

Professional Experience

Toyota Motor Corporation May 2026–Present
Mechanical Design Engineer — Body Electronics Design Division *Toyota, Japan*
- Assigned through CBS Techno. Design sensor-mounting components and packaging in CATIA V5; developed a macro to assess sensor coverage, obstruction, and interference.

Nakayama Iron Works / ITB deLabo
Product Development & Design Engineer

ITB deLabo — Part-time *July 2023–August 2025 | Bandung, Indonesia*
- Designed heavy-machinery components and micro-hydropower layouts; performed structural/flow analyses and prepared manufacturing drawings.

Onsite Internship *September 2024–February 2025 | Takeo, Japan*
- Designed robot mechanisms and evaluated motion, structural strength, and actuator loads using SolidWorks and ANSYS.

Leadership and Community Engagement

Unit Robotika ITB / KRAI | *Head of Mechanical Division / Project Manager* July 2022–June 2024
- Led competition-robot design, fabrication, and integration; contributed to the team's third-place regional result.

Vorione Bladeless Wind Turbine | *Mechanical Lead* December 2023–July 2024
- Led structural design and vibration analysis; coordinated prototype fabrication with electronics and aerodynamics teams.

Society of Renewable Energy ITB | *Community Development* June 2023–June 2024
- Helped plan and install solar panels and supported community outreach explaining solar energy to local residents.

Technical Skills and Languages

Mechanical Design: CAD, mechanisms, actuators, transmissions, linkages.
Simulation: CFD; static, dynamic, and kinematic analysis.
Control: Motor control, teleoperation; MATLAB/Simulink.
Computational Design: CAD/CAE automation, ML surrogate modeling, multi-objective optimization.
Languages: Indonesian: Native | English: Business level | Japanese: Conversational